

PYTHON PROGRAMMING – SYLLABUS

Introduction

Python is a high-level, interpreted programming language.

History

Python was developed by Guido van Rossum in 1991.

File Extension

Python programs are saved using the .py extension.

Variables

Variables are used to store data values.

Concatenation

Concatenation joins two or more strings.

Tokens & Keywords

Keywords are reserved words with predefined meaning.

Identifiers

Identifiers are names given to variables, functions, or classes.

Data Types

Data types define the type of data stored in variables.

Literals

Literals are fixed constant values.

Special Literals

None, True, and False are special literals in Python.

Collections

Collections store multiple values in a single variable.

Operators

Operators perform operations on variables and values.

Arithmetic Operators

Used for mathematical calculations.

Comparison Operators

Used to compare two values.

Logical Operators

Used to combine conditional statements.

Bitwise Operators

Perform operations on binary numbers.

Membership Operators

Check whether a value exists in a sequence.

Special Operators

Dot and colon operators provide special functionality.

Left & Right Shift

Used to shift bits left or right.

Binary & bin()

bin() converts numbers into binary format.

Length

len() returns number of elements.

Slicing

Extracts a portion of a sequence.

User Input

`input()` accepts user input.

If / Elif / Else

Used for decision making.

Nested If

An if statement inside another if.

Loops

Loops repeat a block of code.

Inner Loop

A loop inside another loop.

List Methods

`append`, `extend`, `clear`, and `pop` modify lists.

Tuple Unpacking

Assigns tuple values to variables.

Split

Splits a string into a list.

Set Operations

`Union` and `intersection` perform set operations.

Datatype Check

`type()` and `isinstance()` check data types.

Dictionary Operations

Used to manage key-value pairs.

Functions

Functions are reusable blocks of code.

Callback Function

A function passed as an argument to another function.

Lambda Function

Anonymous function with a single expression.

Filter, Map, Reduce

Used for functional programming.

Class

Blueprint for creating objects.

Constructor

Initializes object data.

Encapsulation

Wrapping data and methods together.

Inheritance

Child class acquires parent class properties.

Polymorphism

Same method with different behaviors.

Abstraction

Hides internal implementation.

Super

Accesses parent class members.

Private Method

Restricted to class usage.

Static Method

Method independent of object.

Inner Class

Class defined inside another class.

Import

Used to include modules.

Platform

Provides system-related information.

Random

Generates random values.

Math

Provides mathematical functions.

Reduce

Applies a function cumulatively.

File Handling

Used to read and write files.

Async & Await

Used for asynchronous execution.

NumPy

Library for numerical computing.

Pandas

Library for data analysis.

Escape Sequences

Special characters like `\n` and `\t`.